Felix Akwasi Sarpong

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Profile

Graduate researcher in statistics and scientific computing with training in mathematics, machine learning, numerical methods, and statistical modeling. Experience applying R, Python, MATLAB, and computational methods to time-series causality, neuroimaging, optimization, Bayesian analysis, and large-scale data problems. Experienced graduate instructor and teaching assistant in mathematics, statistics, and scientific programming.

Education

Georgia State University — Atlanta, Georgia
Master of Science in Statistics — *Aug 2025 – Present*

Georgia State University — Atlanta, Georgia
Master of Science in Mathematics — *Aug 2023 – May 2025*

- Non-Thesis Project: Hierarchical Matrices

African Institute for Mathematical Sciences (AIMS Ghana) — Accra, Ghana
Master of Science in Mathematical Sciences — *Aug 2021 – Jun 2022*

- Thesis: Detecting Causality from Time Series in a Machine Learning Framework

Kwame Nkrumah University of Science and Technology — Kumasi, Ghana
Bachelor of Science in Statistics — *Sep 2016 – Aug 2020*

- Thesis: Time Series Analysis of Rainfall Pattern in Ashanti Region, Ghana

Research Experience

Machine Learning and Computational Neuroscience Research — AIMS Ghana
2021 – 2022

- Investigated causal relationships in multivariate time series using machine-learning methods and reservoir computing.
- Applied computational methods to resting-state neuroimaging data to study directed effective brain connectivity.
- Contributed to collaborative research that developed into a scientific preprint and COSYNE 2024 work.

Hierarchical Matrices Project — Georgia State University
2024 – 2025

- Studied hierarchical matrix representations for efficient approximation and computation with large structured matrices.
- Developed the project as the non-thesis research component of the M.S. in Mathematics.

Publication

J. Falcó-Roget, A. Onicas, **F. A. Sarpong**, and A. Crimi. "Directed networks and resting-state effective brain connectivity with state-space reconstruction using reservoir computing causality." *bioRxiv*, 2023. doi:10.1101/2023.06.08.544175. Related work presented at COSYNE 2024.

Selected Technical Projects

Statistical Learning and Pattern Recognition — *R*

- Built and evaluated regression and classification workflows using training/validation splits, regularization, model selection, and predictive performance measures.

Bayesian Statistical Computing — *R*

- Implemented Bayesian models and simulation-based inference, including conjugate models, Monte Carlo methods, mixture models, and Gibbs sampling.

Numerical Optimization and Scientific Computing — *R / MATLAB*

- Implemented Golden Section Search, steepest descent, conjugate-gradient, Newton/Levenberg–Marquardt, and constrained optimization methods.

Survival and Longitudinal Data Analysis — *R*

- Applied survival models and longitudinal/biostatistical methods to graduate-level data analyses, including model diagnostics and inference.

Teaching Experience

Graduate Assistant — Aug 2023 – Present
Department of Mathematics and Statistics, Georgia State University — *Atlanta, Georgia*

- Support undergraduate instruction in Pre-Calculus, Calculus, College Algebra, Elementary Statistics, and Financial Mathematics.

- Guide students through assignments and quantitative problem solving; grade coursework and examinations and provide constructive feedback.

Teaching Assistant Sep 2020 – Aug 2021
Department of Mathematics, KNUST *Kumasi, Ghana*

- Supported Mathematical Programming instruction using Python, MATLAB, and R and assisted with numerical-analysis practical sessions.
- Helped students with programming assignments, root-finding, interpolation, and computational problem solving.

Technical Skills

Programming	:	R, Python, MATLAB, SQL
Methods	:	Statistical Modeling, Machine Learning, Time Series, Bayesian Computing, Survival Analysis, Numerical Optimization, Scientific Computing
Tools	:	RStudio, LaTeX, Git/GitHub, Microsoft Excel, Power BI
Languages	:	English (fluent), Asante Twi (fluent)

Certification

Blossom Academy Data Analytics Certification Kwabenya, Ghana
Data Analytics *Jan 2023 – Apr 2023*

Leadership and Service

Kids and Maths Organisation Accra, Ghana
Member *Oct 2017 – Present*